where $\Omega_i = k^{-1/2}\mu_i^{-1/2}$ is the frequency of the ith oscillator. Integrating by parts, we have

$$\int_0^t \Omega_i \sin(\Omega_i(t-s))Q(s)\mathrm{d}s = [\cos(\Omega_i(t-s))Q(s)]_0^t - \int_0^t \cos(\Omega_i(t-s))P(s)\mathrm{d}s$$

$$= Q(t) - \cos(\Omega_i t)Q(0) - \int_0^t \cos(\Omega_i(t-s))P(s)\mathrm{d}s.$$

which leads to

$$q_i(t) - Q(t) = \cos(\Omega_i t)q_i(0) + \Omega_i^{-1}\sin(\Omega_i t)p_i/\mu_i - \cos(\Omega_i t)Q(0)$$
$$- \int_0^t \cos(\Omega_i(t-s))P(s)\mathrm{d}s.$$

Introducing this formula in (6.23), we obtain a reduced system

$$\dot{Q} = P, \tag{6.27}$$

$$\dot{P} = -U'(Q) - k^{-1}\sum_{i=1}^{k}(Q - q_i), \tag{6.28}$$

$$= -U'(Q) + f(t) - \int_0^t \psi(t-s)P(s)\mathrm{d}s, \tag{6.29}$$

where we define

$$f(t) = -k^{-1}\sum_{i=1}^{k}\left[\cos(\Omega_i t)q_i(0) + \Omega_i^{-1}\sin(\Omega_i t)p_i(0)/\mu_i - \cos(\Omega_i t)Q(0)\right],$$

$$\psi(t) = k^{-1}\sum_{i=1}^{k}\cos(\Omega_i(t)).$$

For large k and for many choices of the frequencies Ω_i and the initial conditions $\{q_i(0), p_i(0)\}$, function $f(t)$ will take on the character of a Wiener process; in fact the convergence can be shown in the weak sense. We may even take the initial conditions to be randomly distributed [13, 205]. The function $\psi(t)$ will typically exhibit an initial decay followed by rapid bounded fluctuation (see Fig. 6.4), with decay to zero in the limit of large k. Let $\{\Omega_i^{(k)} | i = 1, 2, \ldots, k\}$ represent the frequencies used for a given k. We may make various proposals for the $\Omega_i^{(k)}$,

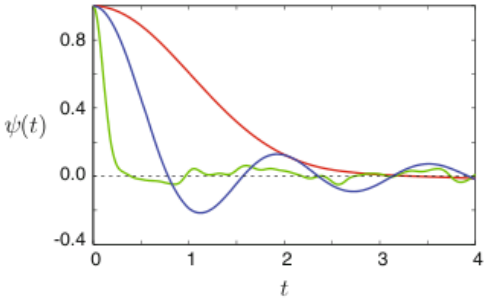


Fig. 6.4 The memory kernel ψ is shown for three different sets of frequencies Ω_i. *Blue*: $k = 400$, $\Omega_i = 0.01i$; *red*: $k = 1{,}000$, $\Omega_i \sim \mathcal{U}[0,1]$; *green*: $k = 1{,}000$, $\Omega_i = 10\hat{\Omega}_i$, where $\hat{\Omega}_i \sim \mathcal{U}[0,1]$

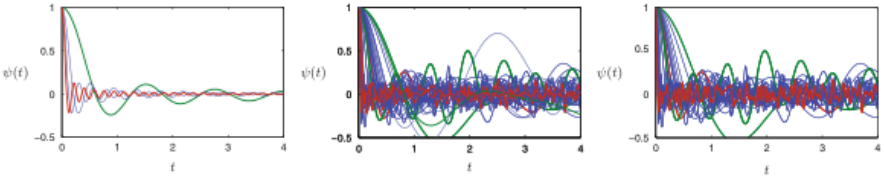


Fig. 6.5 The memory kernels are shown for various choices of $\Omega_i^{(k)}$. *Left*: $\Omega_i^{(k)} = i/\sqrt{k}$, $k = 5^2, 10^2, \ldots, 35^2$; *center*: $\Omega_i = \sqrt{k}\hat{\Omega}_i$, $\hat{\Omega}_i \sim \mathcal{U}[0,1]$, $k = 10, 20, \ldots, 100$; *right*: $\Omega_i = k\hat{\Omega}_i$, $\hat{\Omega}_i \sim \mathcal{U}[0,1]$, $k = 10, 20, \ldots, 100$. In each case the *red curve* indicates the largest value of k considered, whereas *green* indicates the smallest value

for example choosing them to be equally spaced on $[0, k]$, uniformly (randomly) distributed on a fixed interval, or uniformly distributed in an interval of length $\sqrt{k}$. Different choices of sequences $\Omega_i^{(k)}$ will lead to different appearances of the kernel ψ, as shown in Fig. 6.5.

If sufficient high frequency components are present, then the mass of the memory kernel is typically found to be localized around 0; in such a situation of rapid decay it is found that one may replace the function ψ by a Dirac delta function scaled by a positive coefficient γ; we say that the system is Markovian or memory-less in this situation and we replace the delay-differential equation (6.29) by the SDEs

$$\mathrm{d}Q = P\mathrm{d}t, \tag{6.30}$$

$$\mathrm{d}P = -U'(Q)\mathrm{d}t + \sigma\mathrm{dW} - \gamma P\mathrm{d}t, \tag{6.31}$$

where we have viewed the function $f(t)$ as being approximated by a Wiener increment. This system is referred to as Langevin dynamics. The details of the rigorous derivation of (6.30)–(6.31) are beyond the scope of this book; consult the following [13, 113, 138, 150, 185, 198, 205, 350, 402]. We will consider computations of *generalized Langevin dynamics* in Chap. 8. For the moment we